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DELHI-NCR, GHAZIABAD, UP (INDIA)

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Project Report: Heart Disease Analysis using Tableau

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Class – CSIT, 4-C

1. PROJECT TITLE

Heart Disease Analysis using Tableau

2. PROJECT DESCRIPTION

- System Overview: A Tableau-based analytical system for visualizing heart disease risk factors.
- Problem Domain: Cardiovascular health analysis and preventive healthcare.
- Core Technology Used: Tableau, CSV datasets, data cleaning tools.
- Key Capabilities: Interactive dashboards, comparative analysis, storytelling with data.
- Target Users: Healthcare professionals, researchers, students.
- Real-world Relevance: Helps identify lifestyle and demographic factors contributing to heart disease.

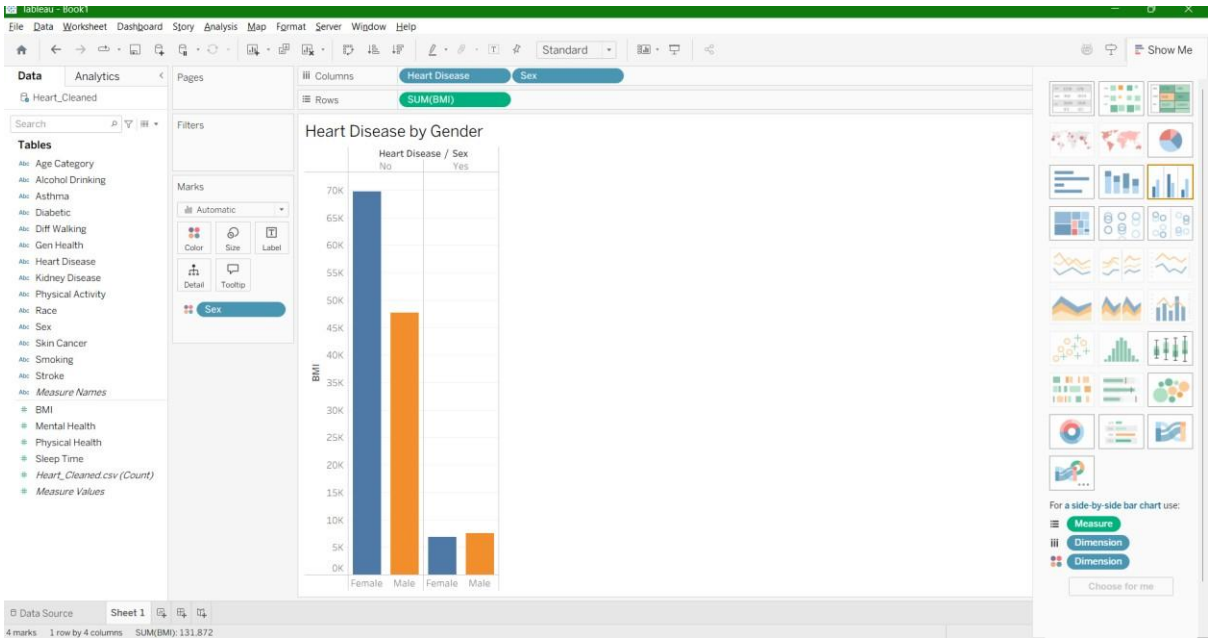
3. APPLICATION SCENARIOS / USE CASES

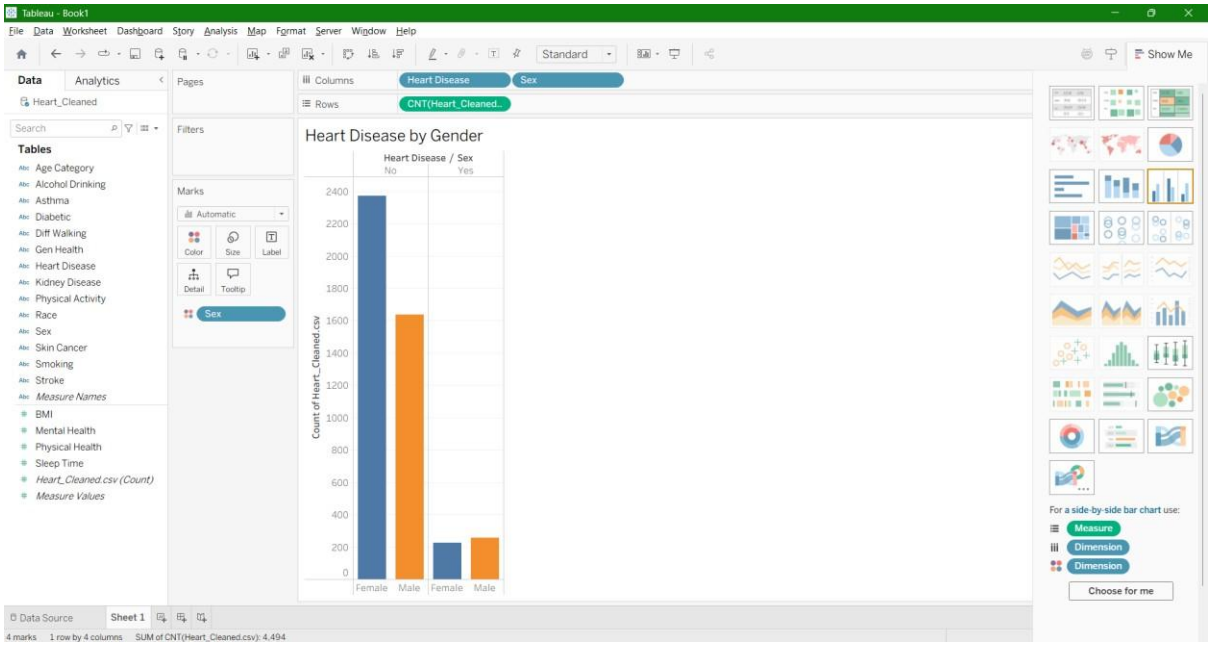
- Healthcare Institutions: Use dashboards to monitor patient risk factors.
- Educational Usage: Teaching students about data visualization and health analytics.
- Research: Supporting studies on lifestyle impacts on cardiovascular health.
- Public Awareness: Visual stories for campaigns on preventive healthcare.

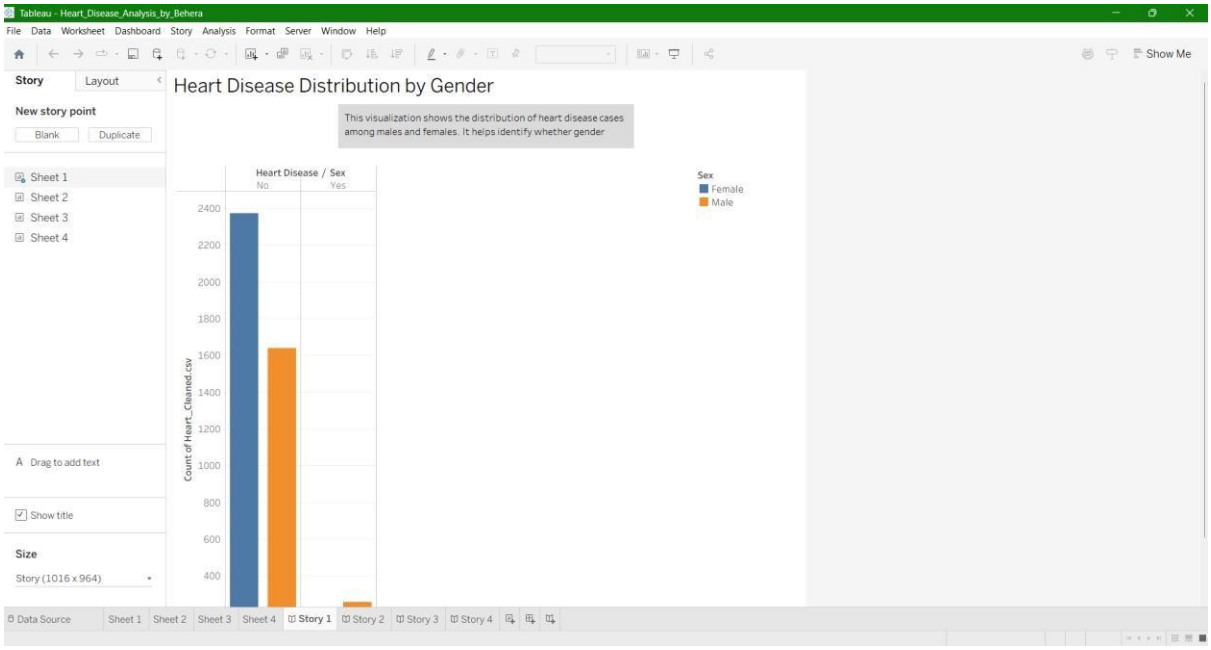
4. TECHNICAL ARCHITECTURE OVERVIEW

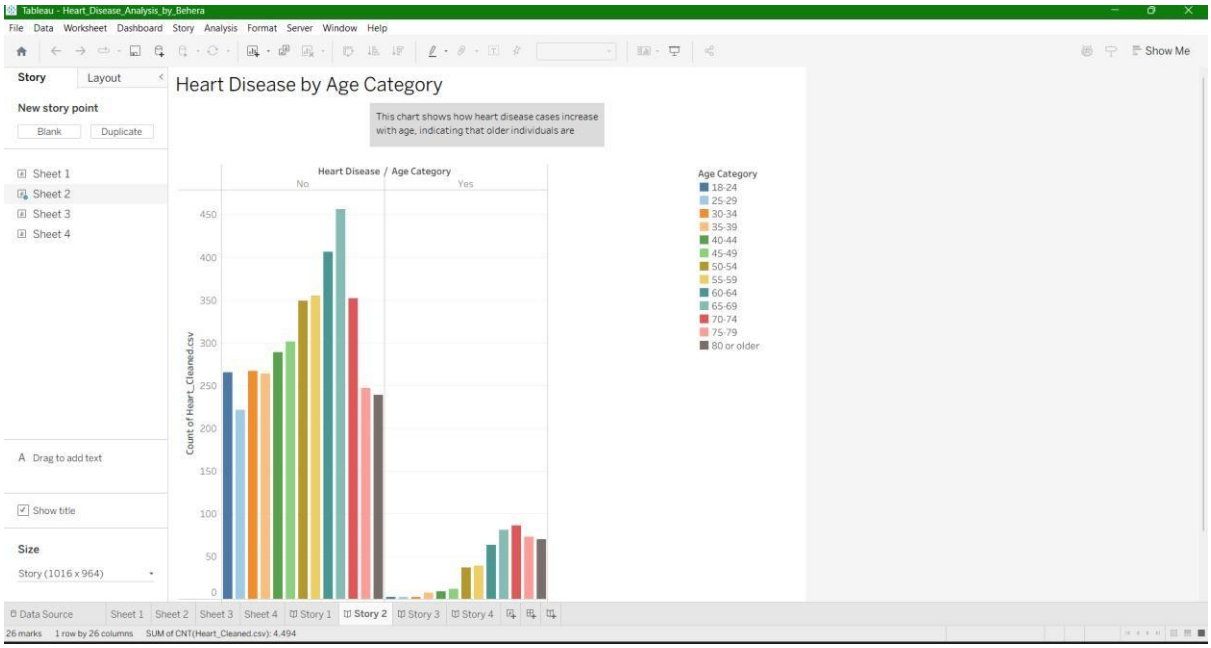
- Frontend Layer: Tableau dashboards and storyboards.
- Backend Layer: CSV dataset preprocessing.
- Database Layer: Heart_Cleaned.csv file.
- External APIs/Services: None.

- **Deployment Environment: Tableau Desktop / Tableau Public.**





- 

- 

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5. PREREQUISITES

5.1 Software Requirements

- **Tableau Desktop / Tableau Public**
- **Python (optional for preprocessing)**
- **GitHub for version control**

5.2 Libraries / Frameworks / Dependencies

- **Pandas (for preprocessing)**
- **Tableau built-in visualization tools**

5.3 Hardware Requirements

- **Standard PC/Laptop with 8GB RAM, 2GHz processor**

6. PRIOR KNOWLEDGE REQUIRED

- **Basics of Tableau**
- **Data cleaning and preprocessing**
- **CSV handling**
- **Visualization principles**

7. PROJECT OBJECTIVES

- **Technical Objectives: Build interactive dashboards.**
- **Performance Objectives: Ensure smooth visualization with large datasets.**
- **Deployment Objectives: Publish dashboards on Tableau Public.**
- **Learning Outcomes: Gain expertise in Tableau storytelling and health data analysis.**

8. SYSTEM WORKFLOW

1. **User loads dataset into Tableau.**
2. **Input handling: Cleaned CSV imported.**
3. **Processing logic: Apply filters, aggregations.**
4. **Core execution: Generate dashboards.**
5. **Output generation: Charts, graphs, storyboards.**
6. **Response delivery: Export reports and share dashboards.**

9. **MILESTONE 1: REQUIREMENT ANALYSIS & SYSTEM DESIGN**
   - **Problem Definition: Identify risk factors of heart disease.**
   - **Functional Requirements: Import dataset, visualize variables, create dashboards.**
   - **Non-Functional Requirements: Scalability, usability, clarity.**
   - **System Design Decisions: Tableau chosen for visualization.**
   - **Technology Stack: Tableau + CSV dataset.**

10. **MILESTONE 2: ENVIRONMENT SETUP & INITIAL CONFIGURATION**
   - **Development Environment Setup: Tableau Desktop.**
   - **Dependency Installation: Pandas (optional).**
   - **Project Structure Creation: GitHub repo with dataset + Tableau files.**
   - **Configuration Setup: Folder structure for dataset, dashboards, reports**

11. **MILESTONE 3: CORE SYSTEM DEVELOPMENT**
   - **Feature 1: Age & Gender Analysis (bar charts).**
   - **Feature 2: Lifestyle factors (smoking, BMI, physical activity).**
   - **Feature 3: Story scenes summarizing findings**

12. **MILESTONE 4: INTEGRATION & OPTIMIZATION** · **Integrated multiple dashboards into a single story.**
   - **Optimized filters for faster response.**
   - **Added validation for dataset consistency.**

13. **MILESTONE 5: TESTING & VALIDATION**

- **Test Cases: Verified dataset import, dashboard rendering.**
- **Unit Testing: CSV preprocessing.**
- **Integration Testing: Tableau story scenes.**
- **User Acceptance Testing: Team reviewed dashboards.**
- **Performance Metrics: Smooth rendering with ~10k records.**

14. DEPLOYMENT

- **Deployment Architecture: Tableau Public.**
- **Hosting Platform: Tableau Public cloud.**
- **Deployment Steps: Upload dashboards, share link.**
- **Production Considerations: Ensure dataset privacy.**

```
project_root/
│
├── dataset/
│   └── Heart_Cleaned.csv
│
├── tableau_files/
│   └── Heart_Disease_Dashboard.twbx
│
├── reports/
│   ├── Behera_Scene_Story-2026.pdf
│   └── Behera_No_of_Visualizations_or_Graphs-2026.pdf
│
├── documentation/
```

```
│   └── Project_Report_DA_Heart_Disease_Analysis.pdf
│
├── images/
│   ├── dashboard_overview.png
│   ├── age_vs_heart_disease.png
│   ├── cholesterol_analysis.png
│   └── chest_pain_analysis.png
│
├── data_processing/
│   └── data_cleaning_notes.txt
│
└── README.md
```

16. RESULTS

- System Output: Interactive dashboards.
- Performance Evaluation: Efficient visualization.

17. ADVANTAGES & LIMITATIONS

- Advantages: Easy visualization, interactive dashboards, realworld relevance.
- Limitations: Limited dataset size, no predictive modeling.

18. FUTURE ENHANCEMENTS

- Add predictive ML models.
- Integrate with larger datasets.

- **Cloud-based deployment.**
- **Mobile-friendly dashboards.**

19. CONCLUSION

This project successfully demonstrates the use of Tableau for analyzing heart disease risk factors. It highlights the importance of lifestyle choices and provides a foundation for further research and healthcare applications.

20. APPENDIX

- **Source Code: GitHub repo**
- **Dataset Details: Heart_Cleaned.csv**
- **GitHub Repository Link: https://github.com/SATYAM8126/Heart-Disease-AnalysisTableau**
- **Demo URL : https://drive.google.com/file/d/1tk2AV3D8tXmquAj8D58453VjFX1knJsc/view?usp=drivesdk**
- **Dashboard URL : https://public.tableau.com/app/profile/vansh.tyagi1985/viz/Heart_Disease_Analysis_Dashboard_17727076550820/Dashboard1**